ENGINEERING PORTFOLIO AERIS
2025 WORLD FINALS

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onshape
ZENO WATCH BASEL
Crédit Mutuel
Rotary
LYCÉE JEAN-MERMOZ
SAINT-LOUIS Agglomération Terres d'avenir
LIONS INTERNATIONAL
VALORIA COMMUNIQUER PAR L'OBJET
ALSACE DÉPANNAGE
Sofitex DE L'INTÉRIM AU CDI
PRODUCTEC LOGICIELS ET SERVICES DE PROGRAMMATION CNC
Biederthal
Ansys
GUERRA FRÈRES Construction
ISL
helbling
SunChemical a member of the DIC group
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INITIAL RESEARCH & TESTING SUMMARY

CAR DESIGN GOAL

Analysis of the engineering scorecards enabled us to define **four goals** to be achieved, which we then ranked according to their **importance**. The car must meet the following criteria:

- **compliant** with technical regulations
- **robust** so that it does not break during races
- **manufacturable** with precision
- **faster** than our previous car at the national finals

SUMMARY OF FORCES APPLIED

Our summary of the forces applied on the {car + CO2 cartridge} system can be broken down into **two types of forces**:

- **driving force, which varies during the race**
- **resistive force, which varies during the race**

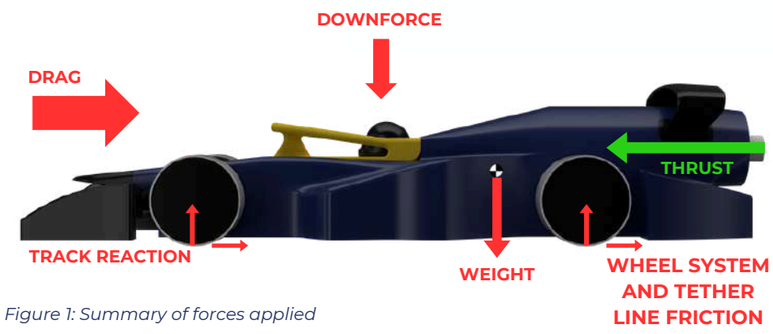


Figure 1: Summary of forces applied

CO2 CARTRIDGE FORCE TESTING AND ANALYSIS.

To measure and analyse the only driving force exerted by the CO_2 cartridge on our car, we designed a **force gauge bench**. This is based on the use of a weighing sensor, two potentiometers and an Arduino board. When the cartridge is pierced, it exerts a **force** on the load cell, causing the steel part to deform. This deformation changes the resistance of the sensor, allowing us to **measure** the force applied.

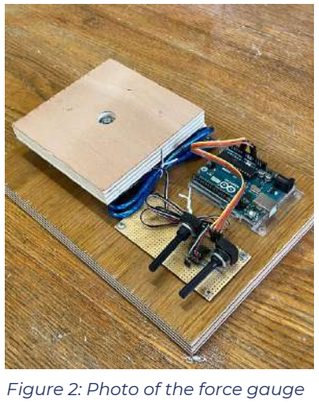


Figure 2: Photo of the force gauge bench used to measure thrust force

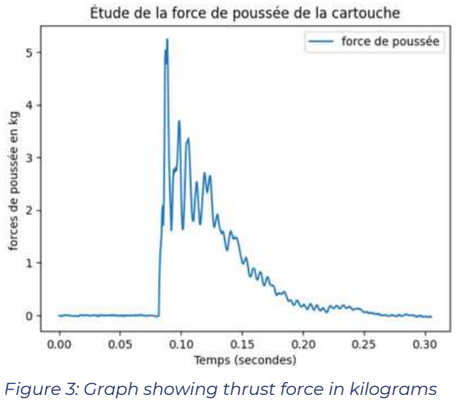


Figure 3: Graph showing thrust force in kilograms versus time in seconds

We then developed a Python programme to collect the data sent by the Arduino, **process it** and then use it. **Aberrant values** are automatically removed. The data is then smoothed and averaged from several trials to improve the **reliability** of the results. Finally, a graph representing the evolution of the average force exerted by the CO_2 cartridge over time is generated, accompanied by a **CSV file** containing the corresponding numerical values.

RACE MODEL DEVELOPEMENT

Using this data, along with information from technical data sheets, simulations and assumptions, we created a **multiphysics model** in MATLAB to estimate the **theoretical track time** of our car. Here's how it works:

```
%% CAR PARAMETERS %%%%
```

CAD TECHNIQUES

CAD SOFTWARE CRITERIA

The choice of CAD software for the STEM Racing project was based on several key criteria :

- Firstly, Onshape's **intuitive interface** allowed us to get up and running quickly: all team members had already been trained in its use over the past three years at high school.
- But above all, Onshape is a **collaborative platform** that allows different **levels of authorisation**. This makes it a perfectly suited tool for this project.

For example:

- The CAD Engineer can build and adjust the geometry in real time.
- The R&D Engineer can access the model to perform simulations, such as CFD.
- The Sponsorship Manager can check the availability of surfaces for adding or validating partner logos.

BRANCH AND VERSION MODELLING

OnShape offers a unique and particularly useful feature: the ability to create **branches and versions** from the same model. This has provided us with several strategic advantages in **managing** our design:

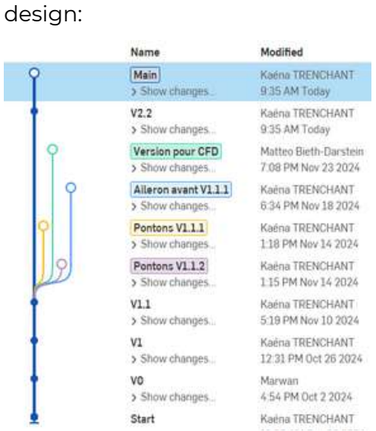


Figure 10: Screenshot of the branches and versions interface in the OnShape software

- **Data security:** each version is permanently saved, allowing us to revert to a previous state at any time without losing data.
- **Parallel work:** we can work on several development axes at the same time without disrupting the main version.
- **Controlled merging:** validated branches can be merged selectively without risk of conflict.
- **Time savings:** everything is done without manual file management, making the process smooth and secure.

USING CONFIGURATION VARIABLES

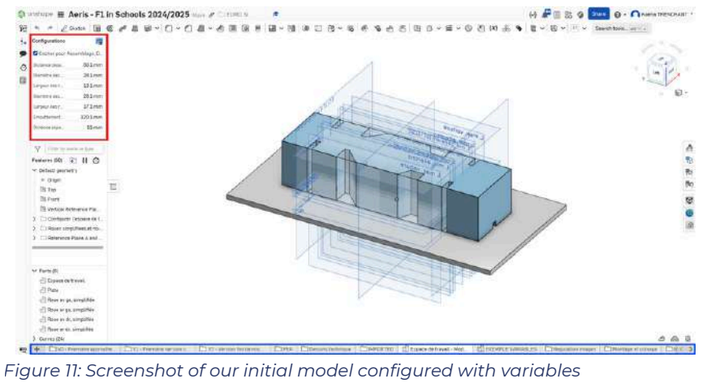
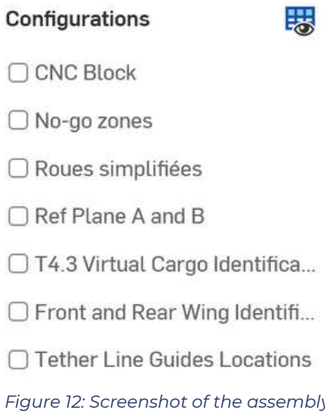


Figure 11: Screenshot of our initial model configured with variables

To **make it easier to modify** to our car and ensure that all dimensions comply with regulations, we used **configuration variables** (red box in figure n*). To do this, all parts had to be modelled in the same space. This allowed us to control the assembly **tolerance between each component** of the car and ensure reliable **geometric continuity** between all parts of the model.



We also used configuration variables in **assemblies** to allow each team member to access a version of the model tailored to their **needs**, without **affecting the work** of others.

For example, the R&D Engineer could tick the “Roues simplifiées” box to run CFD simulations with a **suitable model**. Meanwhile, the Production Engineer could display different configurations to isolate certain components, such as the wings, and generate the corresponding **technical drawings**.

Figure 12: Screenshot of the assembly configuration interface

MODELLING METHODS

OnShape is cloud-based software, which means that **loading times** can be quite long for complex models. To avoid excessive loading times and ensure that the car was **perfectly symmetrical**, we decided to design only half of the car and then apply a mirror feature at the end.

Constrained sketching:

On Onshape, a sketch is said to be constrained when it becomes completely black. To achieve this, we applied a set of **geometric constraints** such as perpendicular, coincident, or vertical.

To ensure geometric continuity between the splines (meeting **G1** and **G2 criteria**), we particularly used the **tangent constraint**.

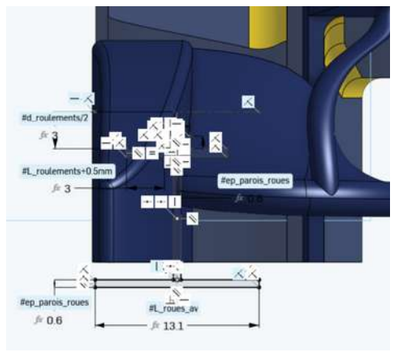


Figure 13: Screenshot of the front wheel sketch

Surface modeling:

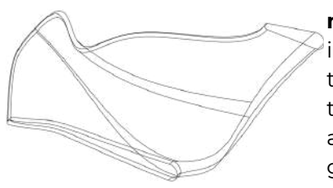


Figure 14: Screenshot of the edges for surface modelling of our front wing

To model the car, particularly the front wing, it was essential to change our method, we had to opt for a **surface modeling method**. This method involves modeling surfaces without thickness, which can then be thickened to obtain complex volumes with precise aerodynamic shapes. This allows for greater freedom in defining **curvatures** and is particularly suited for **smooth transitions** (G1 and G2) where solid modeling reaches its limits.

Variable fillets:

To connect the drop shapes located at the back of the wheels with the cylinder of the cartridge, we used variable fillets. This method allows for the creation of **simpler surfaces** than complex lofts, while maintaining good control over **pressure gradients** and ensuring a smooth and aerodynamic transition.

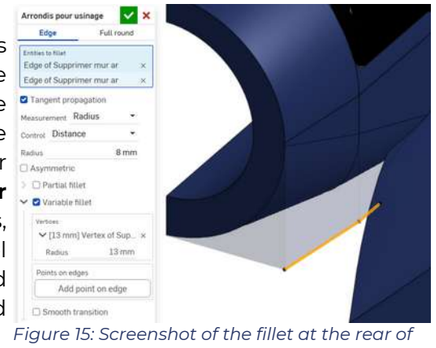
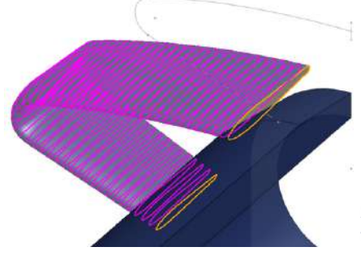


Figure 15: Screenshot of the fillet at the rear of the cartridge chamber

3D splines and multiprofile lofts:



To model more complex shapes, particularly at the rear wing, we used 3D splines with **G1 continuity** at each end and multi-profile lofts. This allows for the creation of smooth and controlled **aerodynamic surfaces**.

Figure 16: Screenshot of the loft for the rear wing

CAD SURFACE ANALYSIS

Zebra analysis:

Zebra analysis allows for visualizing the **continuity between two surfaces**. If the stripes are aligned and continuous, it indicates a good transition (G1 or G2) between the surfaces. We used it to check the **quality of the joints** between separately modeled shapes, such as between the car's nose and the chassis, to avoid gaps that would harm aerodynamics.

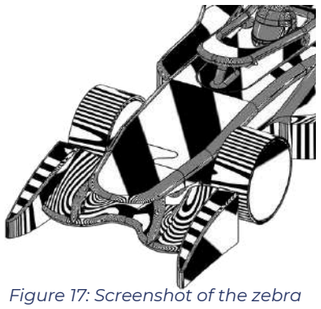


Figure 17: Screenshot of the zebra analysis of the front wing

Interference analysis:

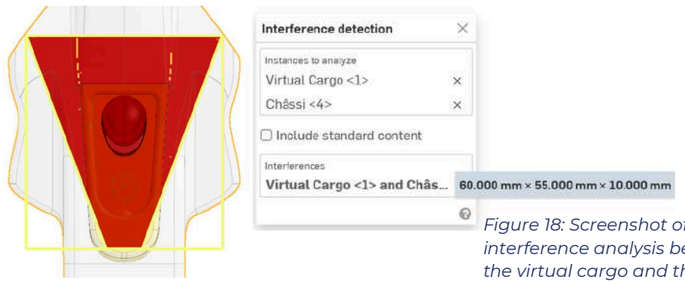


Figure 18: Screenshot of the interference analysis between the virtual cargo and the chassis

The Interference Analysis tool was particularly useful for **validating the integration of the Virtual Cargo** into the chassis: the analysis precisely provided us with the included dimensions, allowing us to confirm that the volume of the virtual cargo was indeed contained within the chassis, an essential condition for complying with the regulations. Moreover, this allowed us to verify that no parts **overlapped** and that each had good assembly tolerance.

MASS OPTIMIZATION

OVERVIEW

Reducing the overall weight of the car is the **main area of development** in our process, especially since regulations have changed from a minimum of 50g to 48g between the national finals and the world finals. Our OnShape modelling software offers an essential feature for this purpose: **live weight tracking**. After selecting the materials for each component, we assigned them on OnShape. This also allowed us to track the **car's centre of mass** and **moment of inertia**.

LIGHTWEIGHT CONCEPTION

To achieve the target weight of 48.1 g, we applied a lightweight design approach. We began by **analysing the densities** of the materials used: we first noticed that nylon was much denser than the foam in the chassis. Our strategy was therefore to transfer volume from the nylon parts to the chassis, for example by minimising the length of the axles.



Figure 19: V1 of the front axle



Figure 20: V3 of the front axle

Next, we focused our efforts on **reducing the volume of nylon parts** such as axles. To do this, we refined certain parts of the components, such as the **thickness** of the part in contact with the chassis. Similarly, we minimised the volume of the rear wing and its attachment **despite its aerodynamic properties**, as the wing does not provide significant downforce for straight-line racing.



Figure 21: V5 of the rear wing

Finally, the most effective strategy was to create strategic recesses. For example, **hollowing** out the car under the sidepods allowed us to reduce the mass of our car by 3.07g. In total, thanks to the hollows under the rear wheel arches, under the sidepods and under the front wing, we were able to **reduce the mass of our car by 7.42g**, or **nearly 13%**.



Figure 22: View of the hollows under the sidepods



Figure 23: View of the hollows under the front wing

ALIGNING THE CENTRE OF GRAVITY

To optimize our performance, it's not just about reducing the mass of our car between 48.1g and 48.3g. For this, it is also necessary to consider the **alignment of the center of gravity** (COG) with the axis of the CO2 cartridge. This helps minimize energy losses. Indeed, the misalignment of the car's COG induces a **moment** relative to the thrust force. However, the formula that allows us to calculate this moment is: Moment (τ) = F × r, where F is the force and r is the **perpendicular distance** from the center of gravity. The objective is to have the value as close to 0 as possible.

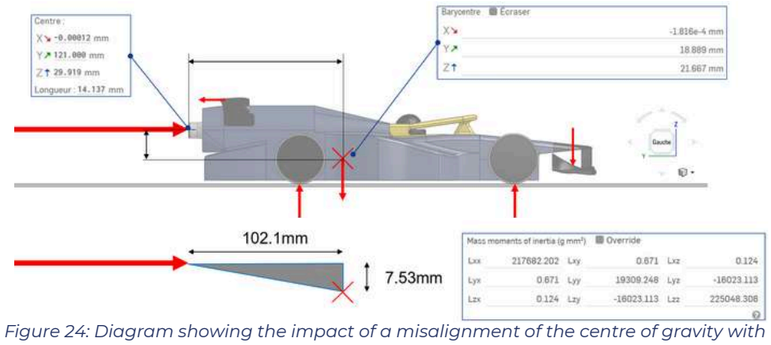


Figure 24: Diagram showing the impact of a misalignment of the centre of gravity with the direction of the thrust force

CHOICE OF MATERIALS

The choice of materials for components such as wings and axles is a point that should not be overlooked during mass optimization, as it represents nearly **a quarter of the car's total mass**.

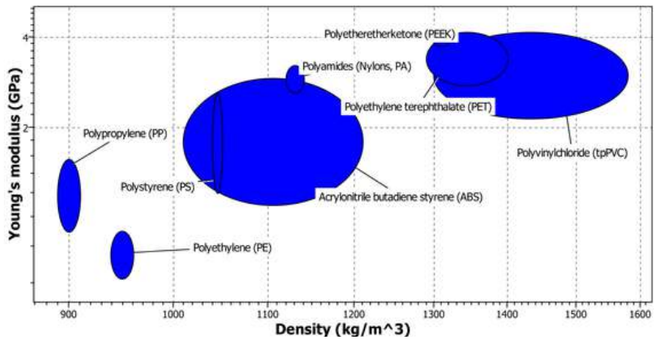


Figure 25: Graph showing the Young's modulus of polymers versus their density

To do this, we used **Granta Edupack**, the software from one of our partners: Ansys. Initially, we considered materials commonly used in 3D printing: PLA, ABS, and PETG. We were familiar with the use of these materials and they had the advantage of being **inexpensive**, but they had to be discarded due to their **low mechanical strength**.

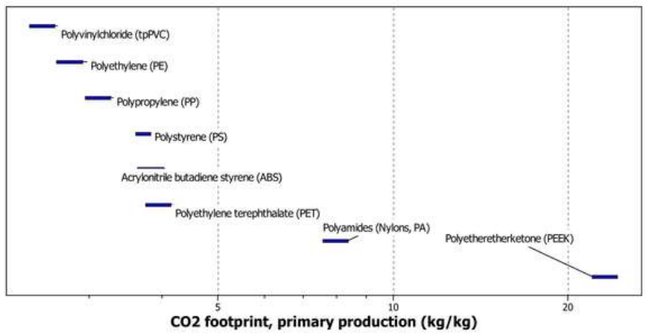


Figure 26: Graph showing the carbon footprint of polymers in kg/kg

In addition to technical performance, we also evaluated the **environmental impact** of these materials. Nylon and PET are relatively energy-intensive polymers to produce, but they offer good durability and recyclability, unlike PEEK, which is more difficult to recycle due to its complex chemical structure.

PP and PE, although more environmentally friendly in terms of carbon footprint, were not compatible with our **mechanical requirements**. Finally, we chose nylon, which offered the best **compromise** between rigidity and reasonable cost. Moreover, it is easily 3D printable, which allows us to produce **precise parts** with good reproducibility.

MASS DISTRIBUTION

We created a mass distribution diagram to visually **identify** the areas that contribute most to the car's total weight. This tool allowed us to target the parts where a mass reduction would be most **effective**. For example, the analysis highlighted that the wings represented a significant portion of the mass, which prompted us to **optimize their geometry** to lighten them.



Figure 27: Circular diagram of the distribution of the car's mass

CHOICE OF PAINT

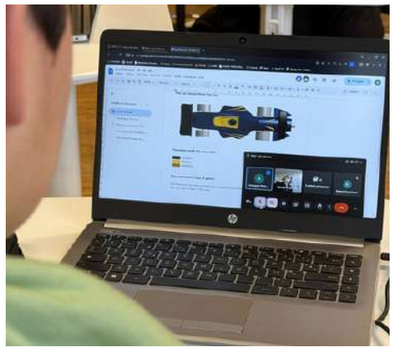


Figure 28: Photo of our video conference with SunChemicals

Thanks to our partner SunChemicals, we were able to paint the car precisely and lightly. After several exchanges of emails and phone calls, we started by thinking about the **livery of our car** with the art direction department of our team, taking into account the **constraints imposed**. We agreed with them that the design of the livery should remain simple and that only the chassis should be painted.

The other parts, such as the front wing, rear wing, axles, flanges, halo and helmet, **retained their original colour**: black. Like Mercedes with the W14's carbon livery, this has enabled us to minimise the weight of the paintwork.

We asked for the use of a hydroxylated functional **acrylic resin**, cross-linked with a polyisocyanate. This process is commonly used in the aviation and automotive industries. The result is a **thin**, high-gloss **coating** with excellent mechanical properties and **resistance** to external conditions and chemicals.

UNCERTAINTY AND ADJUSTMENT

The halo and helmet **are bulky parts** compared to the fins, but because of the regulations they cannot be modified. The loads on these parts are not very high. This is why we **opted for rPLA**, so that we could adjust the mass of the car after painting, by **adjusting the filling of** the halo and helmet. The aim was to find the **best compromise between mass and resistance**.

CFD TECHNIQUES

A PROFESSIONAL COLLABORATION

Mass and its distribution play a crucial role in our car's performance. However, it was inconceivable for us to overlook **aerodynamics** and its effects in the development process. Aerodynamics, though fundamental, remained a field we had not yet had the opportunity to explore in depth. Moreover, this discipline requires **considerable computational resources** to carry out accurate simulations on our model. To overcome this lack of expertise, we sought the help of former students from our Euro SI section, notably Baptiste Masse, who is currently pursuing a PhD **specializing in aerodynamics**. Thanks to his extensive experience in this field and with his **knowledge of fluid mechanics**, he enabled us, through an enriching exchange, to gain a better understanding of the fundamental principles of applied aerodynamics as well as numerical simulation tools, particularly CFD software.



Figure 29: 16/10/2024, Jean Mermoz High School

CFD SOFTWARES

Tool	Main Use	Resources Required	Computation Time	Outcome
Ansys Discovery	Fast parametric studies & sensitivity	1 GPU (workstation)	Minutes	Rapid design iterations
Ansys Fluent / OpenFOAM	High-fidelity aerodynamic validation	HPC cluster (64–512 CPU cores)	Days to weeks (1000× longer)	Accurate drag & downforce predictions

Figure 30: Comparison table of CFD software

This comparison highlights why we deliberately combined the strengths of both approaches. On the one hand, **Ansys Discovery** enabled us to carry out parametric studies in just a **few minutes** on a single GPU, drastically shortening design cycles and allowing us to quickly identify aerodynamic sensitivity zones. On the other hand, **Ansys Fluent and OpenFOAM** provided the necessary **high-fidelity validation**, relying on HPC clusters of 64 to 512 CPU cores and **running for days or even weeks** to accurately capture complex aerodynamic phenomena such as drag and downforce.

By reusing the outputs from **Discovery as a pre-analysis** stage and then transferring the most promising configurations into Fluent and OpenFOAM, we created a **cross-integrated workflow**. This hybrid approach allowed us to save **valuable computation time without sacrificing accuracy**, leveraging the speed of GPU-accelerated solvers while ensuring robust validation with CPU-based high-fidelity simulations. Ultimately, this synergy between tools was key to optimizing our aerodynamic development process.

SIMULATION SETTINGS

Solver Configuration:

- Inlet condition : 20 m/s
- Outlet condition : 0 Pa
- Moving wall :
- Ground : horizontal motion with the same velocity as the airflow
- Wheels : angular velocity defined by the relation:

$$\omega = \frac{vinlet}{rwheel}$$

- Turbulence Model: k-ω SST

Justification of CFD parameters:

1. Incompressibility assumption:

- The **maximum velocity** of the car (V = 25 m/s) corresponds to a Mach number of about 0.073. This **justifies the use of an incompressible solver**, in accordance with the convention that flows can be considered incompressible when the Mach number is below 0.3.
- The Mach number calculation is as follows:

$$M = \frac{V}{c} = \frac{25}{343.3} \approx 0.073$$

Where $c = 343.3 m/s$ is the speed of sound at 20 °C, and $V = 25 m/s$ is the vehicle velocity.

- This result confirms that the flow is incompressible, making this type of solver appropriate.

2. Steady-state assumption:

- Simulations are carried out in a **steady-state regime** to reduce computation time.
- At relatively low velocities (below 25 m/s), **unsteady flow effects remain limited**, which justifies the use of a steady-state solver to obtain reliable results.
- The **k-ω SST turbulence model** was selected for **its accuracy** in capturing separated flow regions and recirculation zones. This model offers a good **compromise between precision and computational cost**.

→ Another model, such as **k-ε**, could have been used, but it is **less accurate** in separated flows.

→ **RANS** models, although less precise than LES models, remain suitable here because LES would **require significantly higher computational costs.**

4. Mesh

- Meshing is a **critical element in CFD simulations**, as it has a major impact on both result accuracy and computation time.
- A simplified wind tunnel environment was adopted around the car, using the **Cartesian Boxes method** (nearfield and farfield regions).

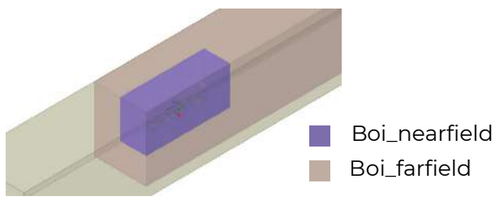


Figure 31: Cartesian Boxes

This methodology relies on a **hierarchical subdivision of the simulation** domain into meshing regions adapted to their relative importance.
The **zones close to the car**, where the velocity and pressure gradients are most significant, are **meshed with a very high density** in order to ensure an accurate capture of aerodynamic phenomena. Conversely, the regions far from the vehicle, where the gradients are less pronounced, use a **coarser mesh**, thus limiting the computational cost.

3D MODEL ADAPTED TO CFD SIMULATIONS

CFD simulations **require very high computational power**. For example, the simulation of our V2 car required the use of a large number of calculation cores for several days. To avoid excessively small time steps and prevent the extension of calculation times, it is **essential to simplify the 3D model** before meshing. To do so, we simplified several parts of the geometry to avoid separation issues. We can observe this on the nose (see Figures n and m).

We can also observe this on the front wing (see Figures 32 and 33)

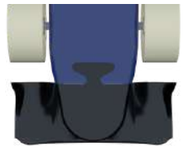


Figure 32: Front wing, V2



Figure 33: Front wing, CFD version

Likewise, the **gap between the halo and the chassis** had to be **removed**. We also made modifications to the **halo** by removing the certification logo of the V2 halo. **Indeed, this created refinement problems** in the mesh and had little interest in the simulation results.

Figure 43 - Zoom on he halo, V2

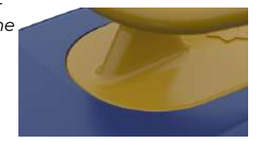




Figure 34: Zoom on the halo, CFD version

We also replaced our real wheels with a **simplified model**, as the turbulence around the wheels required more mesh density and longer calculation times, which prevented us from obtaining significant improvements to our conclusions. Finally, in the V1 model, we also **decided to round the sharp edges to obtain a cleaner mesh.**



Figure 35: Wheel, CFD version



Figure 36: Reel Wheel, V2

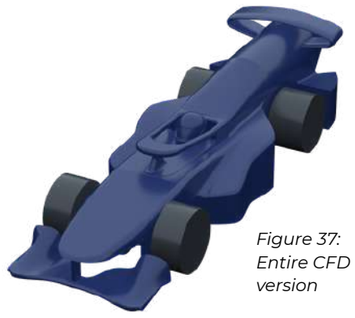


Figure 37: Entire CFD version

ITERATIVE OPTIMISATION

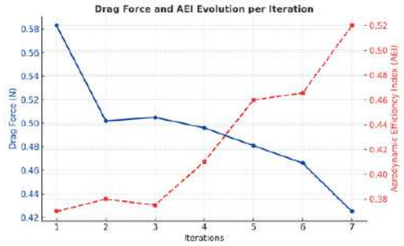


Figure 38: Evolution of drag over the iterations

Aerodynamic Development Methodology

Iteration-Based Framework

Our aerodynamic process followed a strict **iteration-driven approach**, where each iteration marked a fundamental redesign rather than incremental tuning.

- **Major Iterations = Concept Shifts**

A **new iteration** was defined only by a **geometric redesign** of the **chassis**, requiring new CNC machining. This ensured each step delivered a genuine leap in performance.

- Minor Adjustments = Refinements

Smaller changes, such as front wing profiles, rear wing adjustments, or ground linkages, were treated as refinements within **the same iteration**, aimed at optimizing flow coherence and reducing induced drag.

Aerodynamic Efficiency Index

During the development of the sidepods a critical area located between the front and rear wheels our goal was not only to reduce drag but also to avoid aerodynamic instabilities. Excessive lift in this region can make the car unstable, leading to unwanted friction through sliding or oscillations. To account for this, we introduced an improved Aerodynamic Efficiency Index (AEI):

$$AEI = \frac{1}{C_d + 0.3\,|C_l| + 2\,\sigma_{C_l} + 0.5\,\sigma_{C_d}}$$

This enhanced AEI allowed us to **prioritize drag reduction**, while also **penalizing** configurations that generated **significant lift or strong unsteady aerodynamic** loads. For the sidepods, this ensured a design that minimized drag without compromising stability.

FRONT WING DEVELOPMENTS :

While the **regulations required** the integration of a **front wing**, we questioned its true aerodynamic value. **Was it** genuinely **improving performance**, or was it simply a structural element imposed by the rules? To address this, we carried out a baseline simulation on the front wheel alone, isolating its drag without any additional components. This reference allowed us to objectively measure the front wing's real impact and to determine whether it represented a regulatory obligation or a meaningful aerodynamic advantage.

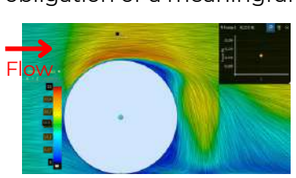


Figure 39 - Control simulation of a single wheel

Drag force : 0.221 N

Simulation on the front wheel alone resulted in a drag **force of 0.221 N**.

FIRST FRONT WING CONCEPT :

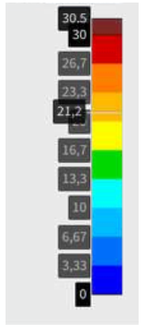




Figure 40: Isometric view, Front wing, First iteration

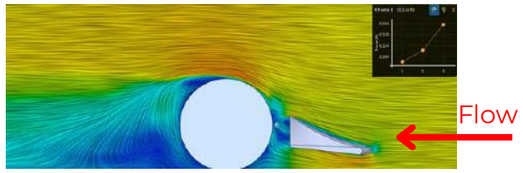


Figure 41: Front Wing version 1 Simulation Drag Force : 0.124 N

With the **addition of the front wing**, drag was reduced to **0.124 N**, a 43.9% decrease compared to the baseline. This proved the front wing as a genuine performance gain rather than just a regulatory constraint.

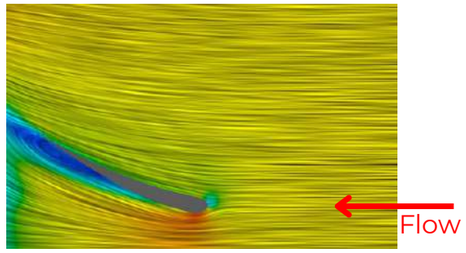


Figure 42: Side view, Air velocity distribution at 20 m/s

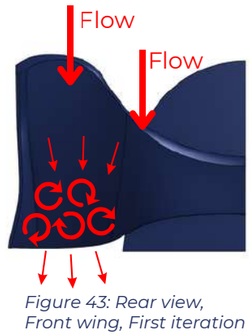
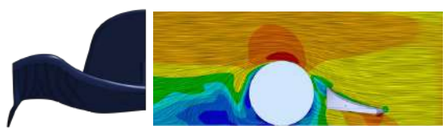


Figure 43: Rear view, Front wing, First iteration

Flow analysis highlighted a **recirculation zone on the intrados**, causing local energy losses.

INTRADOS GEOMETRY OPTIMISATION :



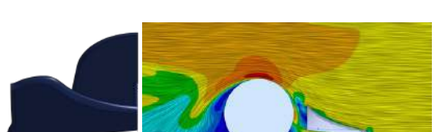
Drag force : 0.126 N

Grooved Intrados: The initial approach introduced grooves to guide the flow and reduce recirculation.



Drag force : 0.121 N

Separated Sections: A second concept split the central and outer parts of the wing, limiting the recirculation surface and encouraging flow attachment.



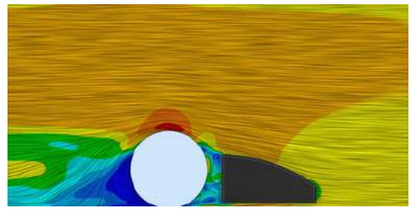
Drag force : 0.101 N

Flat Intrados Plate: Finally, a horizontal plate replaced the curved intrados, completely eliminating the recirculation region.

FINAL FRONT WING DESIGN :



Drag force : 0.062 N

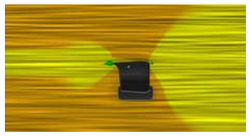


Adapted to World Finals regulations, this design kept the same philosophy as previous versions while pushing the geometry to the regulatory limits for maximum performance.

REAR WING DESIGN APPROACH :



The **rear wing**, **required by regulation**, has only a **limited aerodynamic influence** on our car. Recognizing this, our design approach was to **minimize its drawbacks** rather than seek performance gains. Its geometry was shaped as a streamlined airfoil to **reduce drag** and ensure minimal disturbance of the flow. At the same time, its dimensions were defined at the regulatory minimum, which **reduced both aerodynamic resistance and structural mass.**



This decision aligned with our overall design philosophy: prioritizing weight reduction as the most effective lever for improving acceleration and efficiency. By treating the rear wing not as a performance driver but as a regulatory requirement to be optimized, we were able to integrate it in the most efficient way possible within our car's design.

REAR POD DEVELOPMENTS :

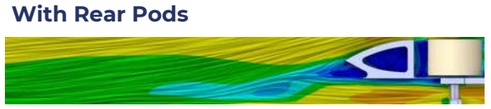
During the **aerodynamic analysis** of our car, we identified a **significant recirculation zone** upstream of the **rear wheels**. This recirculation, caused by the rotation of the wheels, results in higher drag and a noticeable loss of performance. To **address this issue**, we developed **rear pods** positioned downstream of the rear wheels.

Without Rear Pods



Drag force : 0.502 N

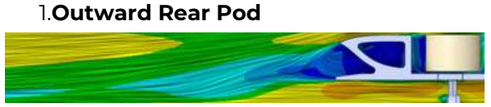
With Rear Pods



Drag force : 0.486 N

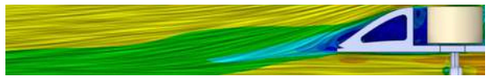
We then explored **three different design concepts** of rear pods, aiming to determine the most effective geometry:

1. **Outward Rear Pod**



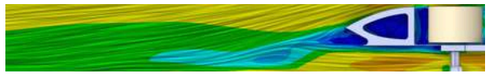
Drag force : 0.504 N

2.**Inward Rear Pod**



Drag force : 0.489 N

3.**Neutral Rear Pod**



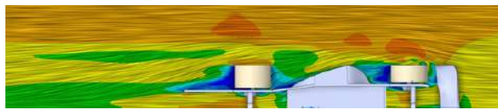
Drag force : 0.486 N

SIDEPOD DEVELOPMENTS

CONCEPT SELECTION

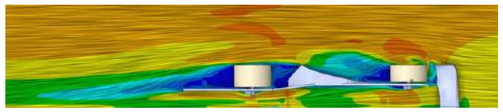
The design of the sidepods had to integrate the virtual cargo, which imposes a certain volume in this region. Three main concepts were evaluated.

Inwash: sidepods shaped like a water droplet, directing the airflow inward.



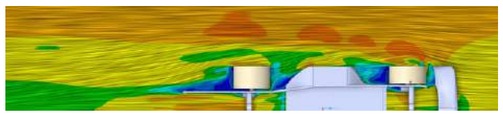
AEI : 0.3751
(Drag Force : 0.505 N)

Outwash: sidepods designed to push the airflow outward, away from the car's centerline.



AEI : 0.3562
(Drag Force : 0.525 N)

Neutral/Wide sidepods: larger, straight profiles with minimal directional effect on the airflow.



AEI : 0.3886
(Drag Force : 0.502 N)

GEOMETRY OPTIMIZATION

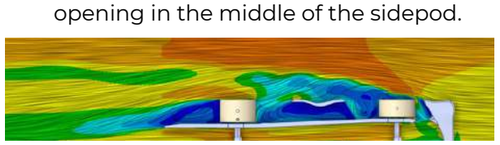
With the chosen concept, **simulations revealed a high-pressure region** at the **front face of the sidepod**, caused by strong recirculation behind the front wheel. This recirculation reduced the quality of the airflow toward the rear of the car.

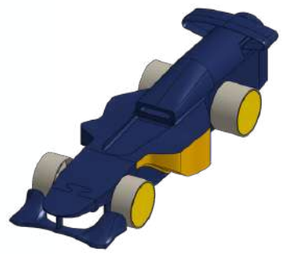
To mitigate this, three design variations were explored:



AEI : 0.3780
(Drag Force : 0.525 N)

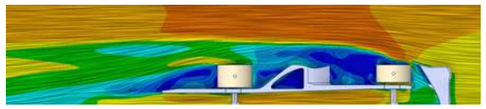
Central perforation: introducing an opening in the middle of the sidepod.

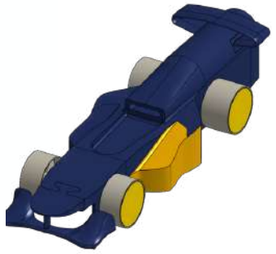




AEI : 0.4367
(Drag Force : 0.501 N)

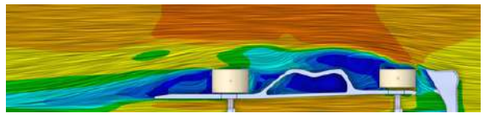
Downward slope: a gradual descending surface, allowing the flow to remain attached and better directed rearward.





AEI : 0.4607
(Drag Force : 0.481 N)

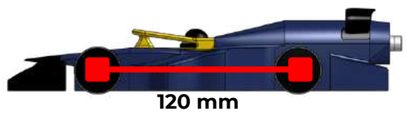
Upward slope: an opposite approach, lifting the flow progressively.



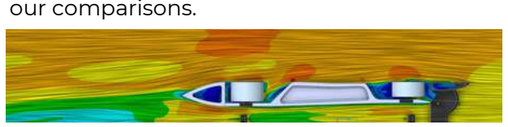
FINAL DEVELOPMENT : TWO-STEP REFINEMENT

The final development phase comprised two complementary improvements aimed at controlling the highly turbulent region between the front and rear wheels and at redirecting the disturbed flow toward the rear of the car.

1. Wheelbase extension increase control surface



We **increased the wheelbase** to enlarge the longitudinal surface area between the front and rear wheels. Rationale: a longer wheelbase provides **more geometric room** to shape and manage the wake from the front wheel, giving the sidepod an extended region to **re-attach and steer the flow rearward**. Practical impact: reduced recirculation intensity in the critical inter-wheel gap and improved robustness of the flow delivered to the rear wheel region, resulting in better aerodynamic stability and a higher AEI in our comparisons.



AEI : 0.4656
(Drag Force : 0.466 N)

2. Duct design : mid-sidepod slit to feed a protected channel

A slit was integrated into the sidepod, channeling a secondary jet of air outward. This jet naturally attached to the ponton surface, **creating a stable side-stream** that prevented separation and improved the quality of the airflow. The result was a **cleaner flow delivered to the rear wheel region**, while also **reducing** the **tunnel contraction effect** that previously generated unwanted vertical forces.



AEI : 0.5247
(Drag Force : 0.425 N)

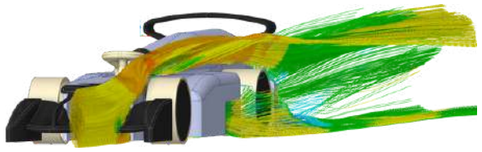


Figure 44: 3D representation of air flow



Figure 45: View from underneath the sidepods

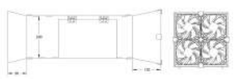
AERODYNAMIC CORRELATION PROCESS

A key challenge in our development process was the **limited accessibility** of **CFD simulations**. Running locally instead of on high-performance servers, our tools **restricted** direct **usage** to only a few engineers. This created a gap: while CFD provided precise numerical insights, **not every member** of the engineer team could explore flow behavior firsthand. To address this, we implemented a **correlation process** between **CFD** and **experimental testing** bringing aerodynamic understanding within reach of the entire team.









The **cornerstone** of this process was the **design** and **construction** of our **own wind tunnel**. Before committing to the final build, we **developed a full-scale cardboard mock-up** to validate its dimensions, flow uniformity, and practicality of use. This iterative step **allowed us** to **refine the design** and confirm that the tunnel would meet our aerodynamic requirements. Once validated, we **built the final version** using **transparent plexiglass**, enabling direct visualization of the airflow and ensuring a robust and durable structure. The **tunnel's dimensions** were carefully chosen to prevent boundary layer interference, while operating conditions were controlled **to keep the Reynolds number low** enough to guarantee laminar flow.

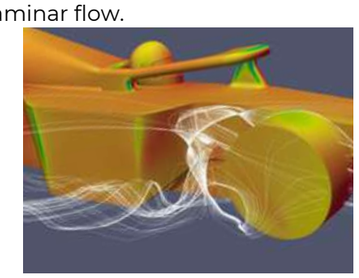


Figure 46: Turbulence behind the front wheels on CFD simulation



Figure 47: Turbulence behind the front wheels in the wind tunnel

While this setup does not allow precise quantitative measurements like CFD, it offers a **unique advantage**: **every engineer** can run a test and immediately **see** the **airflow** in **action**. Separation points, recirculation zones, and wake regions become visible and can be directly compared with CFD predictions. **This combination** of accessibility and physical validation is **what makes the process powerful**. As shown in our images, the **correlation is striking**, the same flow features identified in simulation appear clearly in the tunnel, providing both validation of our digital models and a shared aerodynamic tool for the whole team.

WHEELS SYSTEM & TETHER LINE

OVERVIEW

The wheel system and tether line together constitute our **car's ground connections**. They represent the main source of **non-conservative forces** that oppose movement. Reducing these forces has therefore been a central objective of our research and development efforts.

First, we identified the parameters influencing their intensity, then defined **sub-axes of development** that we ranked according to the expected potential gain. This prioritization allowed us to focus our efforts on solutions offering the best **compromise** between performance and feasibility.

The parameters, ranked from most to least impactful on performance, are:

- **Friction coefficient of the bearings and attachment line guides:** minimize this to limit energy losses due to friction.
- **Moment of inertia and wheel mass:** reduce these values to minimize the energy required to rotate the wheels.
- **Wheel and bearing stiffness:** increase stiffness to reduce the contact surface of the wheels on the ground and within the bearings.

THEORETICAL ANALYSIS OF RACING CONSTRAINTS

Before selecting the bearings, it was essential to verify their **compatibility** with actual operating conditions. We therefore conducted a **theoretical analysis of the specific constraints** involved in STEM Racing competitions. This study enabled us to estimate the maximum rotational speed of the wheels and the mechanical loads exerted on the bearings at the start and throughout the race.

Wheel speed:
Wheel speed is directly linked to vehicle speed and wheel diameter. Taking as a reference a speed of 75 km/h (20.83 m/s) and a wheel diameter of 28 mm, we obtain a **rotation speed** of :

$$P = \pi d \Leftrightarrow P = \pi \times 0.028$$

$$RPS = \frac{V}{P} = \frac{20.83}{\pi \times 0.028}$$

$$RPM = RPS \times 60 = \frac{20.83}{\pi \times 0.028} \approx 14207 rpm$$

The bearings must therefore be capable of operating continuously at a minimum speed of 14,300 rpm, with a **safety margin** to absorb sudden accelerations.

Loads:
The bearings selected must be able to support both axial loads (along the axis of rotation) and radial loads (perpendicular to the axis), without causing **excessive deformation**.

Such deformation would increase the internal and external contact surface, resulting in an **increase in the coefficient of friction** and therefore a **loss of performance**.

Axial load:

$$Fa = m_{total} \times a = 0.0755 \times 1322.5 = 99.84 \approx 100N$$

Radial load:

$$Fr = \frac{n}{m \times g} = 40.050 \times 9.81 = 0.12N$$

BEARING COMPARISON

To maximise speed and **minimise friction**, we **compared the properties** of different types of bearing. This enabled us to make a **pre-selection** by eliminating all the bearings that did not meet our needs or had characteristics that were too weak for our **application**. .

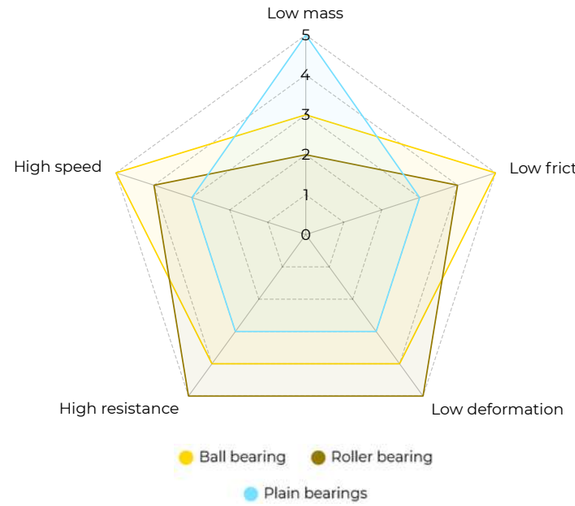


Figure 48: Radar chart comparing bearings

TECHNOLOGY WATCH



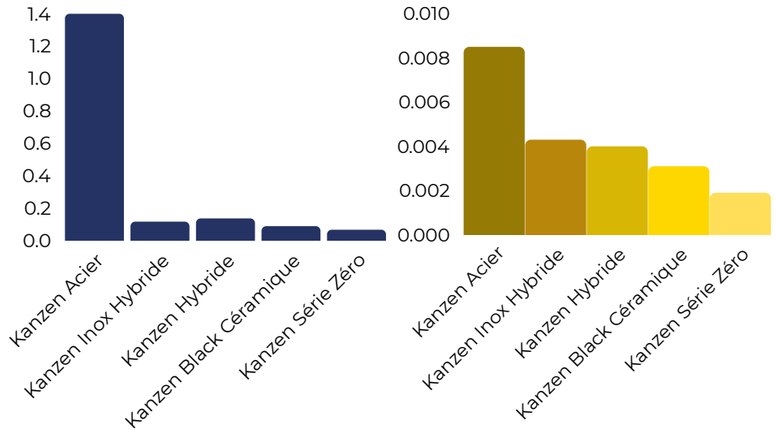
Figure 49: Photo of Kanzen bearings (see website)

We then conducted a technology watch, focusing in particular on **applications using similar processes**. It was during this research that we met our technical partner Kanzen, a company specialising in the manufacture of **hybrid ceramic ball bearings** for fishing rod reels.

These bearings are designed to be compact, achieve **high rotation speeds**, withstand **heavy loads** and maintain a very **low coefficient of friction**.

BALANCING PERFORMANCE AND RESSOURCES

To choose the best bearings, we compared several models, focusing mainly on two criteria: their ability to **reduce friction** and **optimise mechanical efficiency**. The Kanzen Zero Series model would thus offer the best performance with an extremely low coefficient of friction and reduced torque. However, the **delivery time** was too long and the **price** too high, so incompatible with our **Gantt chart**. The Kanzen Black Ceramic, with a coefficient of friction of 0.0031, thus proved to be the **best compromise**.



CHOICE OF TETHER LINES

The attachment guides play an essential role in **safety**, as they must remain perfectly functional throughout the competition. However, they are also a source of **friction**, likely to penalise the performance of our car.



Figure 50: Photo of Igus plain bearings

To limit these **energy losses**, we have chosen to use plain bearings. Using the **comparison tool** provided by Igus, we analysed different models in terms of their coefficient of friction, mechanical strength and durability. The result was the reference **L350SM_0405_04_1**, with the **lowest coefficient of friction** among plain bearings of similar dimensions.

WHEELS GEOMETRY

To maximise performance, wheels must be designed to **minimise the moment of inertia**, while limiting losses due to **ground friction** and **aerodynamic drag**. This is because wheel rotation not only causes friction-generating contact with the ground, but also has an adverse aerodynamic effect: rotation creates low-pressure zones at the rear of the wheels, increasing drag.

WHEELS SYSTEM & TETHER LINE

We therefore began by studying the influence of diameter on these different parameters. A **comparative table** was drawn up, taking into account mass, moment of inertia, ground contact area, frontal area and deformation. The aim was to minimise each of these values to **reduce energy losses**.

Parameters	Large wheels	Small wheels
Mass	Higher	Lower
Moment of inertia	Higher	Lower
Track contact surface	Higher	Lower
Front surface	Higher	Lower
Deformation	Lower	Higher

Figure 51: Comparative table of wheel sizes

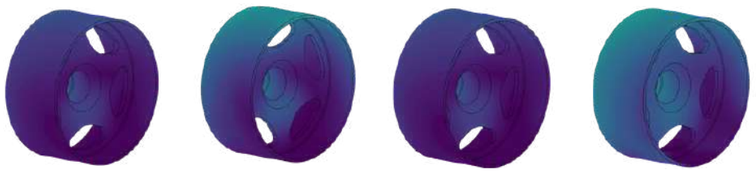
Following this analysis, we opted for a diameter of 28.1 mm, corresponding to the **minimum authorised** by the regulations, with a safety **margin of 0.1 mm** to guarantee compliance during technical inspections.

CHOICE OF WHEEL MATERIALS

The choice of wheel material is decisive in guaranteeing **reliability** and **performance** over all the races. At the national finals, we opted for PEEK wheels, offering a good **compromise** between **mechanical strength and low moment of inertia**. This choice met the requirements of the event, which only required two races, thus limiting the risk of wear or breakage.



Deformation scale in mm



PEEK *TORLON 7130* *Aluminum 7075* *PVC*

For the world finals, we opted for 7075 aluminium, a **high-strength alloy** commonly used in aeronautics and motor sport. Its rigidity and strength virtually eliminate **the risk of deformation or breakage**, reducing the likelihood of **penalties** linked to a damaged component. This choice ensures consistent performance over a far greater number of races than is the case in the national finals.

DESIGN OF CENTRAL RECESSES

The wheels are subject to a number of forces that are crucial to take into account.

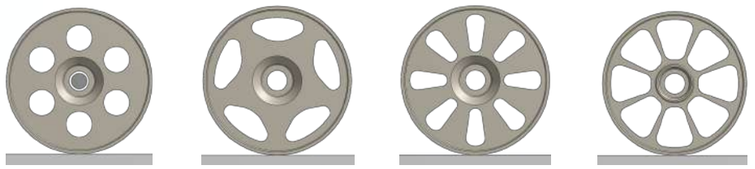
The **vertical load** due to the weight of the vehicle is distributed uniformly over the four wheels and is calculated by:

$$P = m_{total} \times g = 0.074 \times 9.81 = 0.72594N$$

The **wheel safety test** is carried out by suspending a 100 g load perpendicular to each wheel, which generates a force of:

$$F_{safety} = m \times g = 0.1 \times 9.81 = 0.981N$$

The wheels must therefore be capable of withstanding these two stresses. To find the best compromise between mechanical strength and mass, we modelled several wheel designs incorporating central recesses of **different shapes**. For each design, a **finite element analysis** (FEA) was carried out to assess the structural strength.

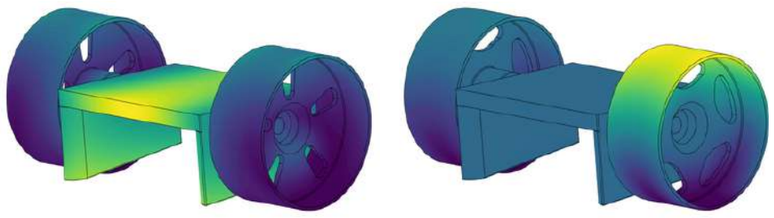


Roue droite V.1.1.5 *Roue droite V.1.1.6* *Roue droite V.1.1.7* *Roue droite V.1.1.8*

The results showed that with the chosen material, **aluminium 7075**, all the designs studied offered a more than **adequate safety margin**. We therefore opted for the **lightest version**, V1.1.8, optimising the **weight/performance ratio** while respecting the mechanical constraints.



Safety factor scale



FEA weight simulation *FEA safety test simulation*

POSITIONING OF BEARINGS

The precise positioning of the bearings is essential to ensure that the wheels **rotate freely** and without **constraint** during the races. At the national finals, **parallelism** problems led to **loss of performance** and **penalties**.

To determine the optimum position, we carried out three finite element analyses (FEA), varying the **number and arrangement** of the bearings. The results showed that the most **stable** and **reliable** configuration was two bearings per wheel.

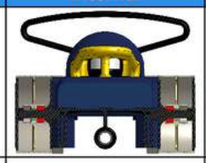
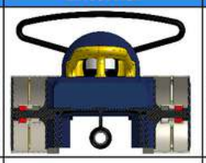
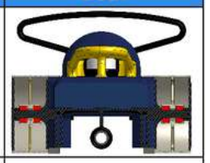
Bearing position	Internal	External	Both
Section view			
Max bearing displacement	3.805e-4 mm	3.812e-4 mm	3.040e-4 mm
Mass	2,80843 g	2,80843 g	3,02543 g

Figure 52: Table comparing the maximum internal displacement of bearings according to their position

GEOMETRY OF THE ESSIEUX



Figure 53: Front axles V2.2

In a conventional system, the axle rotates at the same time as the wheels, resulting in **internal friction** at the bearings. This friction generates **energy losses** that reduce overall performance. Conversely, by opting for a fixed axle and independent wheels rotating on bearings, we significantly **reduce** these losses.

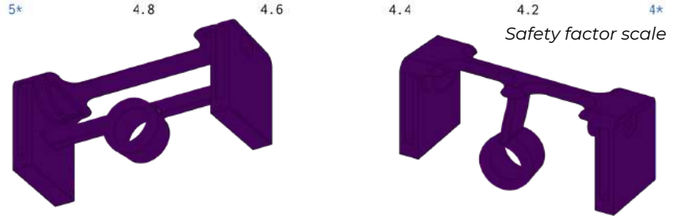
In addition, we have opted for a **discontinuous axis**, meaning that the rotation of the right wheel will have no impact on the rotation of the left wheel. This **avoids amplifying a possible phase shift** in the rotation of the wheels. Finally, it also has an **aerodynamic advantage**, as it slightly reduces the frontal area.

Tether line guides play an essential role in **race safety**, as they hold the car in place on the track using a nylon thread, preventing any risk of it running off or colliding with the environment. They must therefore be able to withstand, **without breaking or deforming**, the repeated **dynamic stresses** generated by acceleration, vibrations and any shocks during starts and finishes.

In order to guarantee this **safety**, we modelled and tested two distinct concepts. The first used two thin, teardrop-shaped side bars designed to reduce drag while providing symmetrical support. The second was based on a single thicker bar positioned at the top of the cylinder.



Safety factor scale



First iteration *Second iteration*

Between the first and second iterations, the weight has been reduced by **4.4%** (from 1,1862 g to 1,1339 g), the **frontal area has been reduced** and the minimum safety factor has been increased by **a factor of 5**.

PRODUCTION, FINISHING & ASSEMBLY

OVERVIEW

The production, finishing and assembly of the car is one of the final stages of the project, and undoubtedly one of the most decisive. This is when our work comes to fruition. The rigour we apply to this phase is crucial, as it determines whether the car **complies with the technical regulations**, is faithful to the 3D model designed and, ultimately, its **performance** on the track.

To guarantee the quality of this stage, we have defined precise criteria for checking **compliance with technical constraints**, such as dimensional compliance with the regulations, **fidelity** to the 3D model, **compliance with machining tolerances**, **mass distribution**, **surface quality**, **compatibility between parts**, the strength of fixings and the durability of finishes.



Figure 54: Photo of the ISL reception area

Planning also played a central role: in collaboration with our partners and suppliers, we set **clear deadlines**, taking into account their **availability** and **delivery times**.

Production of our cars for the finals began on 7 July. It began with the **turning of the wheels** and **machining of the chassis** at our technical partner ISL, a bi-national research institute under the auspices of the French Defence Innovation Agency and the German Ministry of Defence. At the same time, we took care of the **logistics** to ensure that all the necessary parts were ready in time for the world trials.

The production of the nylon parts was entrusted to Sculpteo, a French company specialising in **industrial 3D printing** and additive manufacturing, renowned for its **precision** and speed of execution.

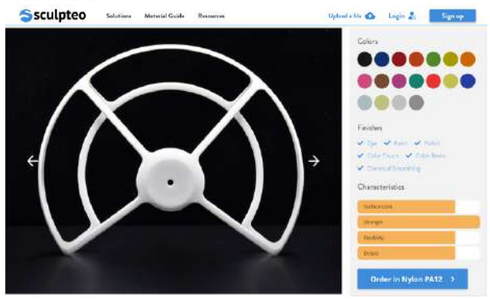


Figure 55: Screenshot of the Sculpteo website, on the Nylon PA12 page

The paintwork was handled by our partner **SunChemicals**, guaranteeing a **professional** and **durable finish**.

The geographical **proximity of ISL**, where the chassis and wheels were produced, gave us the opportunity to follow the process in real time and talk directly to the technicians.

ADDITIVE PRODUCTION OF NYLON PARTS

The small parts of the car, such as the wings and axles, were produced in **PA12 nylon using SLS** (Selective Laser Sintering) 3D printing from our supplier Sculpteo. Unlike the machining of the chassis, we had no direct control over the manufacturing strategy, but we did have to incorporate certain **process constraints** from the design stage onwards, in particular the requirement for a **minimum thickness of 0.8 mm** to guarantee the **feasibility** and **strength** of the parts.



The SLS technique is based on selective laser sintering: a laser beam locally fuses a thin layer of polyamide powder, then a new layer of powder is deposited and solidified in turn.

Figure 56: Checking the front wing for compliance

This **layer-by-layer** process makes it possible to produce complex functional parts without a support, with good **dimensional accuracy** and **mechanical strength** tailored to our needs.

CONSIDERATION OF MACHINABILITY CONSTRAINTS

Even before the design stage, we studied the machining capabilities of our partner ISL, in order to determine the **feasibility** of producing our car. Thanks to their **5-axis machine** and their experience in manufacturing military UAVs in polyurethane, we were able to incorporate **a distinctive side pods shape** that was impossible to achieve on our school's 3-axis machine. This enabled us to realise the **high-performance aerodynamic concepts** we had imagined and modelled without any constraints. What's more, as the smallest milling cutter diameter available is 3 mm, we imposed a **minimum radius of 1.6 mm** on our design to ensure the feasibility of the chassis and wheels.

SUBTRACTIVE CHASSIS PRODUCTION

We came up with an optimised machining strategy, based both on our experience of the national finals and on advice from ISL technicians. At the national finals, the milling cutter was held in a **three-jaw chuck**, which was not precise enough for machining, causing slight offsets and vibrations. For the world finals, we opted for **contraction adjustment** on ISL's 5-axis machine.

The principle is as follows: the **milling cutter is heated by induction**, which slightly expands the metal. It can then be easily inserted into the tool holder.

Once cooled, **the metal contracts** and the cutter is clamped in a very precise and centred manner, guaranteeing **optimum rigidity**. This process reduces the risk of micro-defects associated with holding the tool and improves machining precision.

The Denford polyurethane block was clamped in a vice in a uniform manner, as at the nationals, ensuring optimum stability. The machining strategy comprised **two phases**:

1. Machining the lower part
2. Machining the upper part



Figure 57: First phase of chassis machining

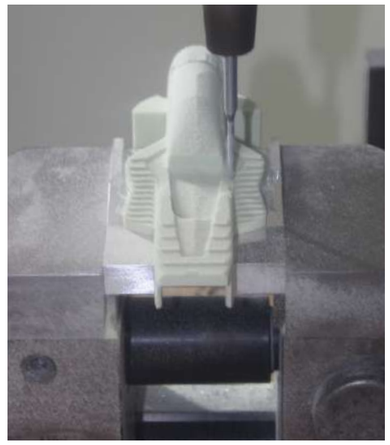


Figure 58: Second phase of chassis machining

To secure the second phase, we proposed to ISL that they make a **flexible aluminium jaw** with a controlled **clearance of 0.05 mm** between the workpiece holder and the workpiece. To prevent **breakage** in fragile areas, we also designed reinforcing **inserts in rPLA** placed in the pontoons (0.1 mm clearance). These parts were modelled very simply using **Boolean operations** and made it possible to stiffen the chassis locally during milling.

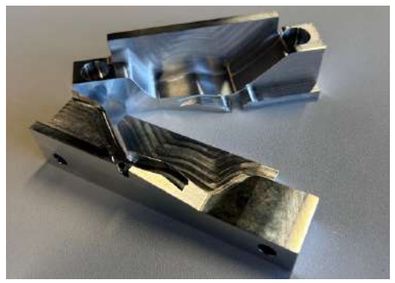


Figure 59: Soft jaws for holding the chassis



Figure 60: rPLA inserts to reinforce the sidepods

The process followed a **progressive logic**: rough milling with a **cylindrical milling cutter** (creating a stepped profile), then finish milling with a **hemispherical helical milling cutter** to obtain the final aerodynamic shape.

PRODUCTION, FINISHING & ASSEMBLY

Unlike the national models, where **manual sanding** had been necessary to correct slight irregularities, the precision achieved with the world models made this step unnecessary.

SUBTRACTIVE WHEELS PRODUCTION

For the world finals, we worked with ISL technicians to come up with a strategy for machining the wheels. Unlike the national finals, we opted for a **7075 aluminium alloy** instead of PEEK, because of its excellent **mechanical properties** and ease of machining. However, we kept to the same method, as it had proved effective in producing wheels that were perfectly within tolerance.

The aluminium bar is fixed in a **three-jaw chuck**. One side of the wheel is machined, then the spokes and bearing housing are milled. We supplied our partner with the bearings and the technical drawing of the wheels. The bearing allows the clearance to be adjusted to a tenth of a millimetre, guaranteeing **precise tolerance** for assembly. The wheel is then cut to separate it from the bar.



Figure 61: Turning the wheels for the national finals

For machining the second side, we designed a jaw with **a clearance of 0.05 mm**, ensuring precise clamping and limiting vibrations. This tried-and-tested approach enabled us to meet the **tolerances specified** in the technical plan.

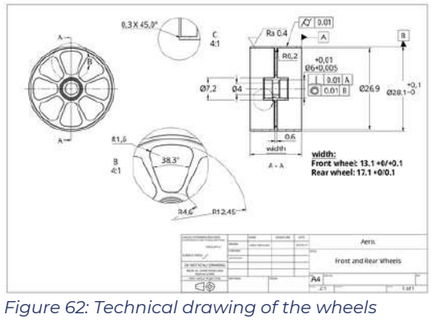
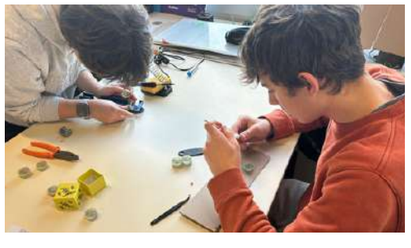


Figure 62: Technical drawing of the wheels



Figure 63: Front wheel of the national finals and its soft jaw

BEARING ASSEMBLY TEST



Fitting the bearings is the most **delicate** and **complex stage** in assembling the car. Good anticipation of the methods to be mastered is therefore essential.

Figure 64: Bearing assembly test

For example, during the national finals, we chose to use **glue** to fix the bearings. But before using this process, we carried out several **preliminary tests** to find both the best technique and the most suitable glue (gel, liquid, etc.). To do this, we used bearings printed in rPLA and wooden rods of the same diameter as the axles.

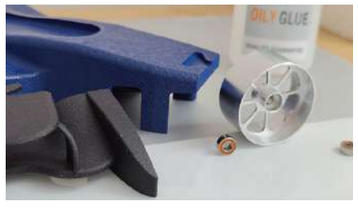
During final assembly, we encountered several problems. Firstly, the gluing operation proved to **be very tricky** to carry out with **precision**: for example, we accidentally put glue in one of the bearings, which prevented the wheel from turning freely and led to a **penalty** at the competition for not complying with the rules. We also noticed a **misalignment of the wheels**, leading to problems of **parallelism** and therefore a **loss of performance**.

These findings led us to look for a more **practical alternative** to gluing the bearings, to ensure that the assembly was both precise and durable.

INSTALLATION OF CONNECTIONS ON THE GROUND

The bearings were assembled at **Zeno Watch Basel**, a Swiss company specialising in **watchmaking**. This enabled us to benefit from their expertise in assembling very **precise mechanical parts**.

First, we degreased the bearings. We then applied a drop of **silicone oil** to ensure the bearings ran smoothly while limiting friction. Unlike at the national finals, we decided to keep the **seals** on the bearings, to **protect** them from dust and outside particles, because at the world finals, performance has to be maintained over a **greater number of races**.





ZENO WATCH BASEL

Figure 65: Images taken during our day of press-fitting bearings into the wheels.

The final assembly is based on a press-fit system with very **precise tolerances**, ensuring that the bearings are **locked on** one side by the axles and on the other by the flanges.

This configuration allows the wheels to **turn freely** while maintaining **perfect alignment**, which considerably improves **performance** compared with the national finals.

RISK MANAGMENT

Although most of the parts were manufactured by our partners, we took the time to consider the **risks associated** with the processes used in the project.

Hazard	Likelihood	Severity	Mitigation
Powder inhalation / skin contact from SLS Nylon PA12	3	8	Respirator and protective gloves used when handling powder. Work area kept under extraction, powder residues cleaned after build.
Burns due to hot surface from SLS chamber	2	7	Cooling cycle respected before part extraction. Heat-resistant gloves worn when removing build.
Fumes or filament particles from 3D printing (FDM)	4	6	Printer enclosed with ventilation. Room ventilated, operators avoid direct exposure.
Cuts or eye irritation during polyurethane foam machining (CNC milling)	3	7	Goggles and dust mask worn, vacuum extraction active during machining, hands kept clear of toolpath.
Dust inhalation from polyurethane foam	5	5	Milling performed with extraction system, operators wear respirator. Workspace cleaned after operation.
Lacerations from sharp chips during aluminium 7075 turning	3	9	Protective gloves used when handling chips, chip shield installed on lathe, regular cleaning of work area.
Entanglement risk (hair, clothes, fingers) during lathe operations	2	10	Operators tie back hair, no loose clothing. Only handles touched while lathe is in motion, emergency stop accessible.
High forces / tool breakage during aluminium or PEEK turning	2	8	Proper tool clamping verified, cutting parameters controlled, protective visor worn.
Heat exposure from PEEK turning (high cutting temperatures)	2	6	Coolant applied when necessary, thermal gloves used when handling freshly cut pieces.

FINAL EVALUATION

In conclusion, our engineering approach focused on three key levers.

1. **Reduction of resistive forces:** Aerodynamics were optimized by reducing the frontal area and adopting continuous shapes (C1/C2) to limit drag. Friction was reduced thanks to Kanzen hybrid ceramic bearings (μ = 0.0031) and Igus plain bearings.
2. **Minimization of inertia and mass:** The wheels have been designed with the minimum regulatory diameter of 28.1 mm and a honeycomb design in 7075 aluminum, optimized by FEA iterations to reduce the moment of inertia while maintaining their mechanical strength. The total mass, around 48.3 g, is the result of strategic recesses and the choice of lightweight materials such as PA12 nylon.
3. **Production and compliance:** 5-axis machining, SLS printing, and high-precision assembly resulted in a product faithful to the CAD design, ensuring that the simulated performance was achieved.

The MATLAB/Simulink model allowed us **to monitor the theoretical performance** of our car throughout the development process and ultimately estimate a **track time of 1.067 seconds over 24 meters**.